3 Exploratory Data Analysis (EDA)

This section conducts exploratory data analysis to summarize descriptive characteristics, visualize distributions and relationships, identify patterns and anomalies, and provide initial interpretations.

3.1 Cross-Ticker Comparison

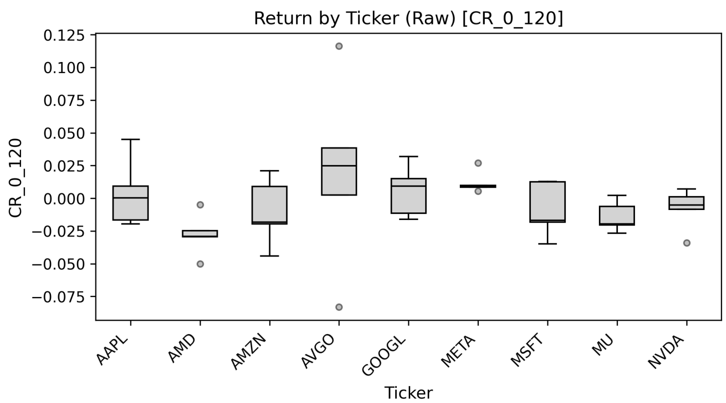

Figure 1: Distribution of cumulative log returns (CR_0_120) across tickers.

Figure 1 presents cumulative log returns by ticker. AVGGO exhibits the highest median return with substantial dispersion, while AMD and MU display predominantly negative medians. AAPL, GOOGL, and META cluster around zero with moderate variability. Several outliers indicate episodic extreme movements, suggesting heterogeneous firm-level performance.

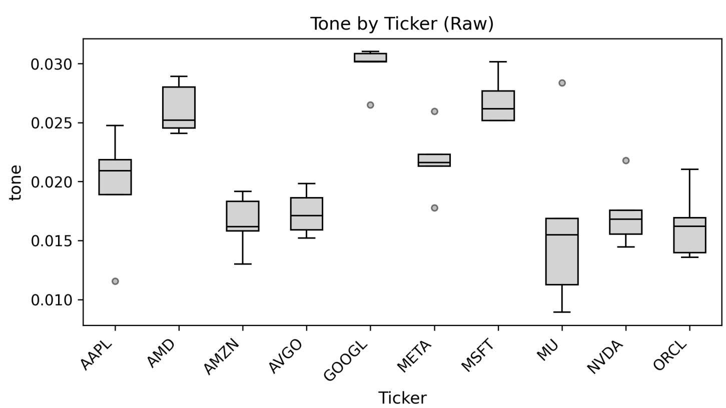

Figure 2: Tone distribution by ticker.

Tone distributions across firms (Figure 2) reveal pronounced cross-sectional variation in narrative sentiment. GOOGL and MSFT exhibit consistently higher median tone levels with relatively tight interquartile ranges, suggesting more stable and positive communication patterns. In contrast, MU and AMZN display lower central tendencies and wider dispersion, indicating greater variability in sentiment across observations.

Notably, several tickers exhibit outliers, implying episodic shifts in tone that may correspond to firm-specific events or earnings-related announcements. Overall, the heterogeneous tone profiles across firms highlight the importance of accounting for cross-sectional effects when assessing sentiment-based signals.

3.2 Distributional Properties

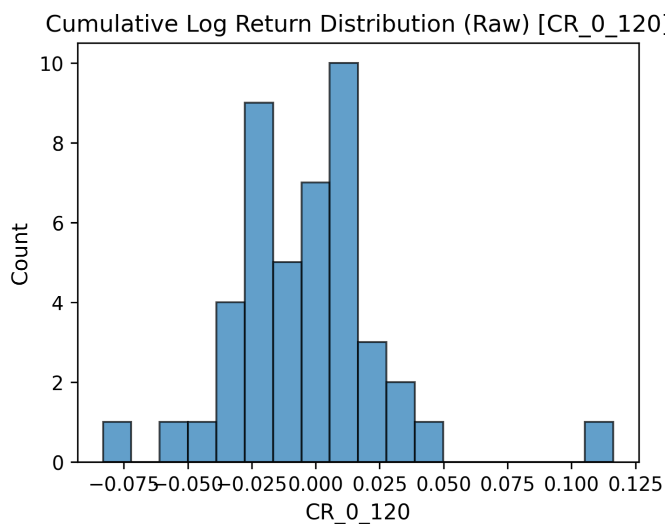

Figure 3: Histogram of cumulative log returns (CR_0_120).

Figure 3 illustrates the distribution of cumulative log returns (CR_0_120). Returns are centered near zero with noticeable tail behavior, indicating the presence of both substantial gains and losses over the observation window. This non-normal shape suggests potential skewness and excess kurtosis, motivating caution when applying linear models that assume Gaussian errors.

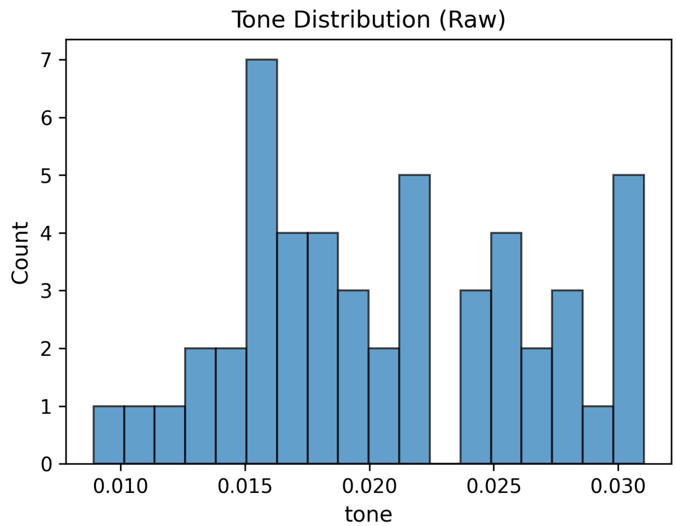

Figure 4: Histogram of tone values.

Tone values shown in Figure 4 are concentrated primarily between 0.015 and 0.025, exhibiting mild right skewness. This clustering indicates that most narrative sentiment remains moderately positive, with relatively few extreme observations.

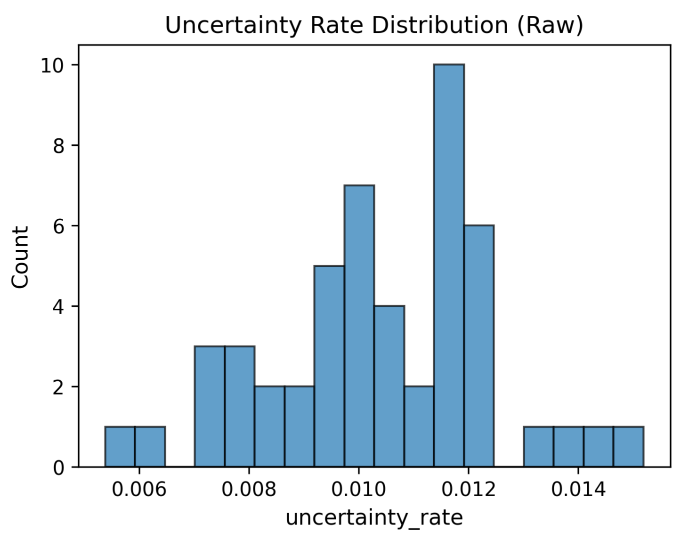

Figure 5: Distribution of uncertainty rate.

The uncertainty rate distribution in Figure 5 appears tightly clustered around its mean, reflecting generally stable uncertainty levels across samples. However, several high-uncertainty outliers are present, which may correspond to periods of heightened informational ambiguity, such as earnings releases or macroeconomic shocks. These distributional characteristics suggest that while tone remains relatively stable, uncertainty exhibits episodic spikes that may carry economic significance.

3.3 Correlation Structure

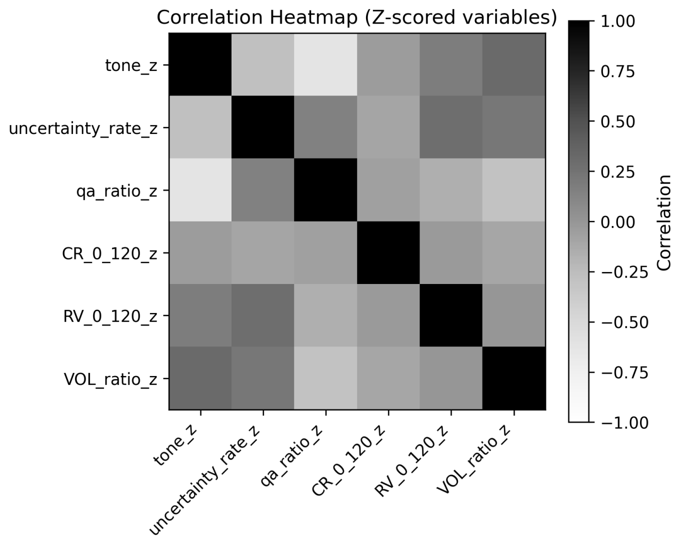

Figure 6: Correlation heatmap of z-scored variables.

The correlation heatmap in Figure 6 suggests generally weak to moderate linear relationships. Tone exhibits only weak association with returns, while uncertainty rate shows a stronger positive correlation with realized volatility. Volume ratio also displays moderate correlation with uncertainty.

3.4 Bivariate Relationships

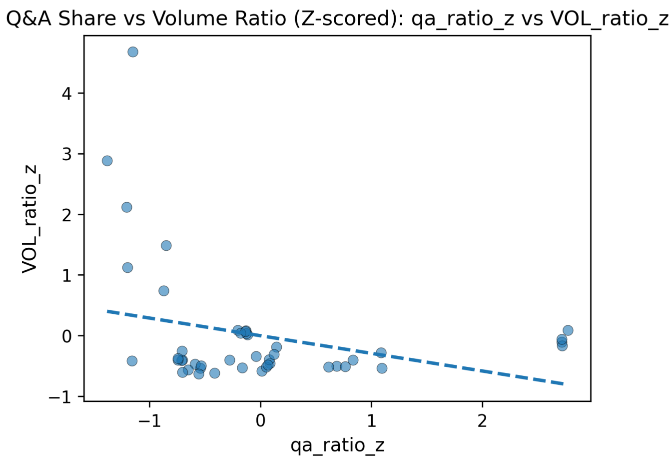

Figure 7: Q&A share versus volume ratio (z-scored).

Figure 7 examines the relationship between Q&A share and trading volume (both z-scored). The fitted regression line indicates a mild negative association, suggesting that higher relative Q&A participation does not necessarily coincide with increased market activity. This result implies that investor engagement during Q&A sessions may reflect information-seeking behavior rather than immediate trading responses.

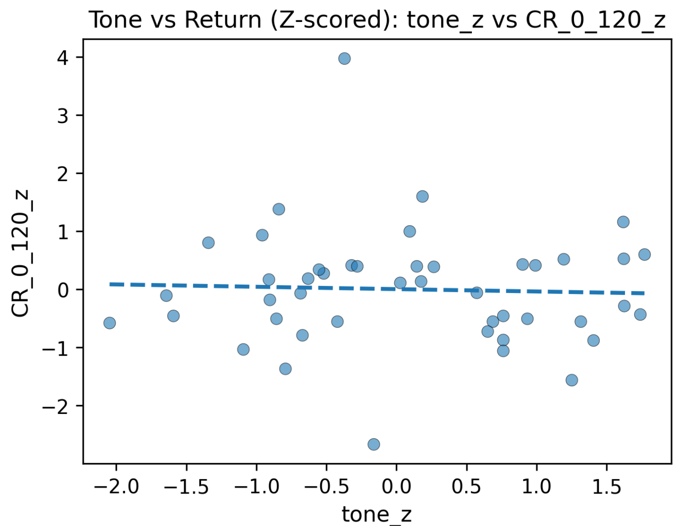

Figure 8: Tone versus cumulative return (z-scored).

Tone versus cumulative return is presented in Figure 8. The near-horizontal fitted line indicates a very weak linear relationship, suggesting that narrative tone alone provides limited explanatory power for short-horizon returns. The wide dispersion of observations further reinforces the noisy nature of sentiment-return dynamics.

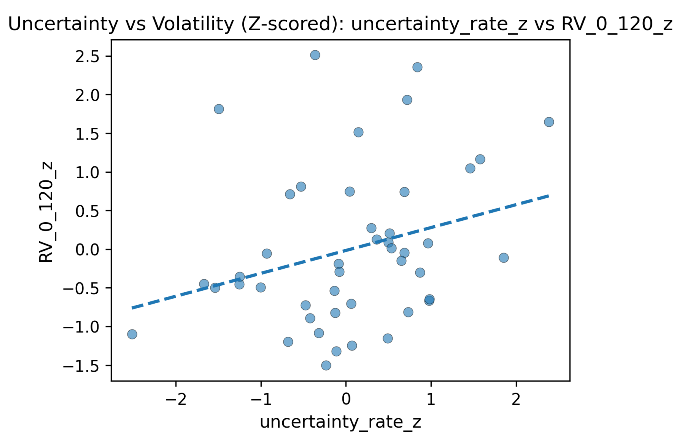

Figure 9: Uncertainty rate versus realized volatility (z-scored).

In contrast, Figure 9 reveals a clear positive relationship between uncertainty rate and realized volatility. Higher uncertainty levels are associated with elevated volatility, consistent with the interpretation that informational ambiguity increases market risk. Compared to tone-based sentiment, uncertainty measures appear to exhibit a more robust and economically meaningful association with market dynamics.

3.5 Initial Interpretations

Overall, sentiment tone appears weakly related to subsequent returns, while uncertainty measures show stronger alignment with volatility dynamics. This suggests that narrative uncertainty may be more informative for assessing market risk than tone-based sentiment. The presence of cross-ticker heterogeneity and outliers highlights the importance of firm-specific effects and motivates further regression analysis in subsequent sections.